Do Now · retrieval
RISC and CISC

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why can instruction fetch and data access contend in Von Neumann architecture?

Q2

What is a pipeline hazard?

Q3

What is an opcode?

Q4

Array [2,4,6,8,10,12,14,16,18], target 16. Give midpoint indexes inspected with low=0 high=8 and integer midpoint.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

RISC approach — Typically favours a smaller set of comparatively simple, regular instructions, often using explicit load/store operations.

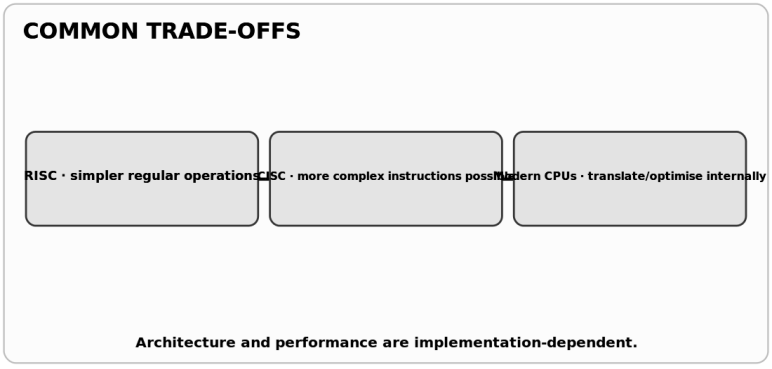
Core idea 2

CISC approach — Can offer complex instructions that express several low-level operations in one instruction, potentially improving code density.

Core idea 3

Avoid oversimplifying — Fewer instructions in assembly do not guarantee fewer clock cycles. Actual performance depends on pipelining, decoding, caching and processor design.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

RISC — Reduced Instruction Set Computer design approach.

CISC — Complex Instruction Set Computer design approach.

load/store — Access memory through dedicated instructions.

code density — Amount of program function per unit of machine-code storage.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

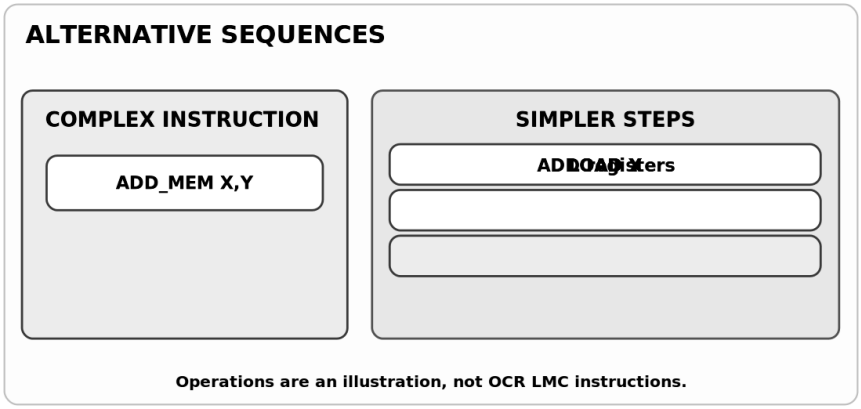
Explain why a load/store architecture needs separate load and add instructions.

Follow the worked method

1. Predict first — Values are brought from memory into registers before the arithmetic instruction.

2. Explain and check — A regular load/store instruction set simplifies certain datapaths but may need more assembly instructions.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
The complex ISA version uses fewer listed instructions but could require more work per instruction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State two characteristics of RISC and two of CISC.

Question 2

Compare one CISC-style memory-add instruction against a RISC-style LOAD, LOAD, ADD, STORE sequence for the same task.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate the claim 'RISC is always faster because its instructions are simpler'. Include code density, decoding and pipelining.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A program written using fewer assembly instructions always executes faster.

Use key vocabulary from memory

1. Explain RISC in your own words.

2. Explain CISC in your own words.

3. Explain load/store in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.